MAT2002 **Name (Print):** _____
Spring 2026
HW2
Deadline: 4th March

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1. Solve the following equation:
   (i) $x^2y^3 + x\left(1+y^2\right)y' = 0$
   (ii) $\left(\frac{\sin y}{y} - 2e^{-x}\sin x\right) + \left(\frac{\cos y + 2e^{-x}\cos x}{y}\right)y' = 0$

2. Show that if $(N_x - M_y)/(xM - yN) = R$, where R depends on the quantity xy only, then the differential equation
$$M + Ny' = 0$$
has an integrating factor of the form $\mu(xy)$. Find a general formula for this integrating factor.

3. Solve $\left(3x + \frac{6}{y}\right) + \left(\frac{x^2}{y} + 3\frac{y}{x}\right)\frac{dy}{dx} = 0$.

4. Determine a formula for the n-th Picard approximation for the initial value problem
$$\frac{\mathrm{d}y}{\mathrm{d}t} = ay, y(0) = 1, a \in \mathbb{R}$$
What is the limiting function $y(t) = \lim_{n\to\infty} y_n(t)$. Is it a solution? Are there other solutions that we may have missed?

5. (i) Find the exact solution of the initial value problem
$$\frac{\mathrm{d}y}{\mathrm{d}t} = y^2, y(0) = 1.$$
   (ii) Calculate the first three Picard approximations $y_1(t), y_2(t)$, and $y_3(t)$ and compare these results with the exact solution.

6. Find the Wronskian of the given pair of functions
   (i) $e^t\sin t, \quad e^t\cos t.$
   (ii) $\cos^2\theta, \quad 1 + \cos 2\theta$

7. For each of the following, determine the longest interval in which the given initial value problem is certain to have a unique twice-differentiable solution. Do not attempt to find the solution.
   (i) $y'' + (\cos t)y' + 3(\ln|t|)y = 0, \quad y(2) = 3, \quad y'(2) = 1.$
   (ii) $(x-3)y'' + xy' + (\ln|x|)y = 0, \quad y(1) = 0, \quad y'(1) = 1$
   (iii) $(x-2)y'' + y' + (x-2)(\tan x)y = 0, \quad y(3) = 1, \quad y'(3) = 2$

8. For each the following, find the Wronskian of two solutions of the given differential equation without solving the equation.
   (i) $(\cos t)y'' + (\sin t)y' - ty = 0$
   (ii) $x^2y'' + xy' + \left(x^2 - v^2\right)y = 0, \quad$ Bessel's equation
   (iii) $\left(1-x^2\right)y'' - 2xy' + \alpha(\alpha+1)y = 0, \quad$ Legendre's equation